

Stationary distribⁿ:

π , a prob on the state space $\mathcal{S}$,

such that $\pi = \pi P$.

$$\boxed{\pi(i) = \sum_{j \in \mathcal{S}} \pi(j) p_{ji}} \quad \text{— balance eqⁿ.}$$

$\forall i \in \mathcal{S},$

If $\boxed{\pi(i) p_{ij} = \pi(j) p_{ji}} \quad \forall i, j \in \mathcal{S}$

then π is a st. distribⁿ $\rightarrow$ detailed balance eqⁿ.

Ex:- RW on graphs, (weights $w_{ij} = w_{ji}$

$$\& w_{i\cdot} = \sum_{j \in S} w_{ij}$$

$$\pi(i) = \frac{w_{i\cdot}}{w_{\cdot\cdot}},$$

satisfies the detailed

$$w_{\cdot\cdot} = \sum_{i \in S} w_{i\cdot})$$

Ex:- Birth & death chain.

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & \dots & N \\ \hline 0 & r_0 & p_0 & & & \\ 1 & q_1 & r_1 & p_1 & & \\ 2 & & q_2 & r_2 & p_2 & \\ \vdots & & & & \ddots & \\ \vdots & & & & & q_N & r_N \end{array}$$

$$\begin{aligned} r_0, r_N &< 1 \\ q_i, p_i &> 0 \\ i &= 1, 2, \dots, N-1 \end{aligned}$$

Sp. $\pi(i) p_{ij} = \pi(j) p_{ji} \quad \forall i \geq 0.$

For $i=0$, $\pi(0) p_{01} = \pi(1) p_{10}$

$\Rightarrow \pi(0) p_0 = \pi(1) q_1. \quad \text{--- ①}$

For $i=N$,

$\pi(N) q_N = \pi(N-1) p_{N-1} \quad \text{--- ②}$

For $i=1, 2, \dots, N-1$,

$\pi(i) p_i = \pi(i+1) q_{i+1} \quad \left| \quad \pi(i) q_i = \pi(i-1) p_{i-1} \right.$

Thus, the eqⁿ are

$$\pi(i) p_i = \pi(i+1) q_{i+1}$$

for $i=0, 1, \dots, N-1$.

Then,
$$\pi(i+1) = \pi(i) \frac{p_i}{q_{i+1}}$$

where,

$$b_{i+1} = \prod_{j=0}^i \frac{p_j}{q_{j+1}}$$

$$b_0 = 1$$

$$\begin{aligned} &= \pi(i-1) \cdot \frac{p_i}{q_{i+1}} \cdot \frac{p_{i-1}}{q_i} \\ &= \pi(0) \frac{p_i}{q_{i+1}} \cdot \frac{p_{i-1}}{q_i} \\ &= \pi(0) b_{i+1} \frac{p_i}{q_{i+1}} \cdot \frac{p_{i-1}}{q_i} \dots \frac{p_0}{q_1} \end{aligned}$$

$$\pi(i) = \pi(0) b_i \quad \checkmark$$

$$\text{Now, } \sum_{j=0}^N \pi(j) = 1.$$

$$\Rightarrow \sum_{j=0}^N \pi(0) b_j = 1.$$

$$\Rightarrow \pi(0) = \frac{1}{\sum_{j=0}^N b_j}.$$

$$\text{Hence, } \pi(i) = \frac{b_i}{\sum_{j=0}^N b_j}.$$

For infinite state sp.

$$\sum_{j=0}^{\infty} \pi(j) = 1$$

$$\Rightarrow \pi(0) \sum_{j=0}^{\infty} b_j = 1.$$

Case 1 :-

$$\sum_{j=0}^{\infty} b_j < \infty, \text{ then } \pi(0) = \frac{1}{\sum_{j=0}^{\infty} b_j}.$$

$$\Rightarrow \pi(j) = \frac{b_j}{\sum_{j=0}^{\infty} b_j}.$$

- There exists a st. distrⁿ.

Case 2 :- If $\sum_{j=0}^{\infty} b_j = \infty$, then $\exists$ no.

solⁿ of $\underline{\pi} = (\pi(i) : i \in S)$

There exists no. st. distrⁿ. Hence the
is not pos. rec.

Ex $P = \begin{pmatrix} q_0 & p_0 \\ q & 0 & p \\ & q & 0 & p \end{pmatrix}$, $q_0 = q$ elastic
bdy
 $q_0 = 0$ is reflecting
bdy.

$$b_j = \prod_{i=0}^{j-1} \frac{p_i}{q_{i+1}} = \frac{p_0}{q^j} p^{j-1}$$

$$\text{Then, } \sum b_j < \infty \text{ iff } \frac{p}{q} \left(\frac{p}{q} \right)^{j-1}$$

$$p < q$$

Ehrenfest model: $\pi(i) = \binom{N}{i} 2^{-N}$

Bernoulli Laplace model: $\pi(i) = \frac{\binom{N}{i}^2}{\binom{2N}{N}}$

Markov Chain Monte Carlo. (MCMC)

Ising Model. V , $\mathcal{S} = \{0, 1\}^V$

$$P(s) \propto \exp(H(s)).$$

$$P(s) = \frac{1}{\textcircled{Z}} \exp(H(s)), \quad Z = \sum_{s \in \mathcal{S}} \exp(H(s)).$$

partition
fn. / 10

Bayesian Model.

$$\theta \in \Theta,$$

$$X_1, \dots, X_n \sim f(x|\theta)$$

$$\theta \sim \pi(\cdot)$$

$$\begin{aligned} P(A_i|B) &= \frac{P(A_i \cap B)}{P(B)} P(B|A_i) = ? \\ &= \frac{P(B|A_i)P(A_i)}{\sum_i P(B|A_i)P(A_i)} \end{aligned}$$
$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\sum_{\psi} f(x|\psi)\pi(\psi)^{\text{prior}}}$$

We want to a sample or even
avg. value from some. distⁿ

$$\pi(i) = \frac{f(i)}{C}, \quad i \in S,$$

Goal: Construct an
irr. M.C. such that
 π is the st. distⁿ of
the chain.

C is const.
unknown/very large.
 $f(i)$ is known
and can be computed.

- $X_0 \sim \mu_0$, μ_0 known. distrⁿ.
 - Run the M.C. suff long, then return the value of X_n at that time as the sample.
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Hastings Metropolis algorithm.

$$X_n = i,$$

$H = ((h_{ij})) \rightarrow$ stochastic
 $\rightarrow$ proposal matrix. matrix.

Consider a sample Y_n with

distrⁿ $Y_n \sim H_i = (h(i, j) : j \in S)$.

$$P(Y_n = j) = h_{ij}$$

$$P(Z_{ij} = 1) = a_{ij}$$

Define, $A = ((a_{ij}))$, $0 \leq a_{ij} \leq 1$, acceptance
prob. matrix.
 $X_{n+1} = \begin{cases} j & \text{if } Y_n = j \text{ \& } Z_{ij} = 1. \\ X_n & \text{o.w.} \end{cases}$

$$p_{ij} = \begin{cases} h_{ij} a_{ij} & j \neq i \\ 1 - \sum_{j \neq i} h_{ij} a_{ij} & j = i \end{cases}$$

Set

$$a_{ij} = \min \left\{ 1, \frac{\pi(i) h_{ij}}{\pi(j) h_{ji}} \right\}.$$

$$= \min \left\{ 1, \frac{f(j) h_{ji}}{f(i) h_{ij}} \right\}.$$

Claim: $\pi = (\pi(i) : i \in S)$ is the st. distrⁿ.

For $i \neq j$,

$$\pi(i) p_{ij} = \frac{f(i)}{C} \times h_{ij} \times a_{ij} = \frac{f(i) h_{ji}}{C}$$

$$\pi(j) p_{ji} = \frac{f(j)}{C} \times h_{ji} \times a_{ji} = \frac{f(j) h_{ji}}{C}$$

$$a_{ij} = \min \left\{ 1, \frac{f(j) h_{ji}}{f(i) h_{ij}} \right\}$$

Case 1 : $f(i) h_{ij} \geq f(j) h_{ji} \Rightarrow a_{ij} = \frac{f(j) h_{ji}}{f(i) h_{ij}}, a_{ji} = 1$

